MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
THIRD SEMESTER EXAMINATION
(2024 ADMISSION ONWARDS)
MG3DSEAQC200 – Seaweed Culture and Utilisation

Duration: 1.5 hrs **Maximum Marks:** 50

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

Part A
Very Short Answer Questions
Answer All Questions
Each question carries 1 mark

1. Name one economically important green seaweed from the class Chlorophyceae. [1] [U]
2. What is the scientific name of *wakame*, a commercially cultivated brown seaweed? [1] [R]
3. Which red algae genus is commonly used in the production of nori? [3] [U]
4. ExpandIMTA in seaweed aquaculture. [2] [R]
5. Name one method of asexual reproduction in seaweeds. [4] [U]
6. What is mannitol, and from which type of seaweed is it extracted? [3] [U]
7. Howis the post-harvest technology in seaweed culture? [4] [U]
8. Give one criterion for selecting a good site for seaweed culture. [2] [E]
9. What is the main component extracted from brown algae like *Laminaria*? [3] [R]
10. Mention one cause of mortality in seaweed culture operations. [2] [U]

[1 x 10 = 10] Marks

Part B
Short Answer Questions
Answer any 5 Questions
Each question carries 6 marks

11. Describe the sexual and asexual reproduction methods in seaweeds. [4] [U]
12. Explain the different types of seaweed culture methods practiced in India. [2] [An]
13. Discuss the criteria for selecting a good site for seaweed culture in open waters and seawater ponds. [2] [U]
14. Write a short note on the distribution of seaweeds along the Indian coast. [3] [U]

15. Outline the post-harvest processing steps for seaweeds, including cleaning, washing, and storage. [3] [R]

16. Write a note on the application of seaweeds as Pharmaceuticals. [3] [U]

17. Explain the types of seaweed culture. [2] [R]

18. Give a taxonomic account on the Green Algae with suitable examples [1] [R]

19. Explain the recent and past developments in seaweed culture operations [2] [U]

[6 x 5 = 30] Marks

Part C

Essay Type Questions

Answer any one Question

Each question carries 10 marks

20. Compare and contrast the taxonomy, morphology, and reproduction of *Enteromorpha compressa*, *Undaria pinnatifida*, and *Porphyra* spp. [1] [U]

21. Explain the chemical composition of seaweeds and discuss the extraction processes for alginates, agar, carrageenan, and mannitol. [3] [R]

22. Discuss the factors affecting the growth of seaweeds, including environmental monitoring, causes of mortality, and recent developments in commercial-scale seaweed culture. [2] [U]

[10 x 1 = 10] Marks